Capacitance Measurement

Capacitance

- Is a measure of the stored electric charge between two objects in an electric field

$$q = CV$$

- Where ***C*** is the capacitance of the capacitor, ***q*** is the amount of stored charge and ***V*** is the potential across the capacitor
- A capacitor consists of two conducting electrodes which are insulated from each other by a dielectric. It stores charge

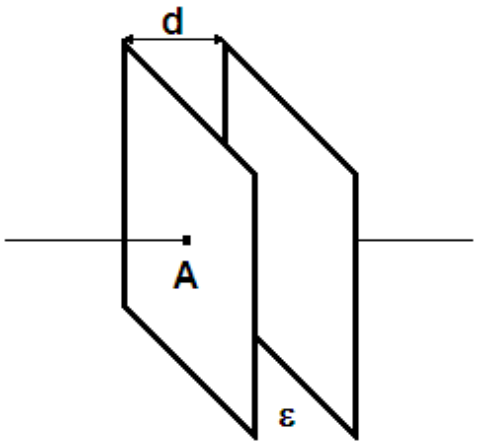


- The capacitance of a conductor depends on its physical geometry and its dielectric properties

- It is measured in Farads. Large unit hence microfarad or picofarad mostly used

Capacitors

- One common example of a capacitor arrangement is the parallel-plate capacitor



- It consists of two parallel electrodes separated by some form of dielectric medium
- The capacitance of this arrangement is:

$$C = \frac{A\varepsilon}{d}$$

Where:

A is the area of one plate,

d is the distance between them

ε is the permittivity of the dielectric material between them

- This relationship does not take into account fringing effects at the edges of the capacitor

Permittivity

- Describes how an electric current effects a dielectric
- It gives and indication of how easily a material polarizes in the presence of an electric potential
- It is defined in terms of the permittivity of free space ε_0 and a relative permittivity of a material ε_r

$$\varepsilon = \varepsilon_0 \varepsilon_r$$

- ε_0 is 8.85 x 10^{-12} F/m

Approximate Values of Permittivity for Common Dielectrics

Vacuum	1
Air	1.0006
Water	~78
Paper	~2.5
Plastics, Oils and Alcohols	~2-6
Glass	~7.5
Ferroelectric Ceramics	~7500

- Permittivity is dependant on temperature and compositional variations, which provides a useful basis for some sensors

Capacitance Measurements

- We can capacitively measure any variable that causes a change in A, d or ε
- Variations in A or d and be measured:
  - Linearly
  - Angularly
- Variations in ε result from a change in the composition between the plates, which could also be temperature dependant

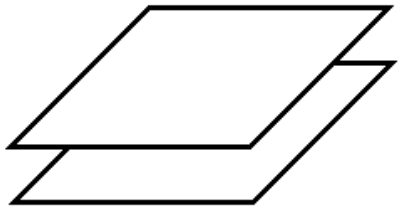
Where Can We Use Capacitance?

- Linear or Angular Motion:
  - Temperature
  - Strain
  - Size
  - Proximity
  - Force
  - Acceleration
  - Vibration → Sound Level
  - Pressure
- Compositional Variation:
  - Depth
  - Moisture Content
  - Continuity
  - Phase Composition

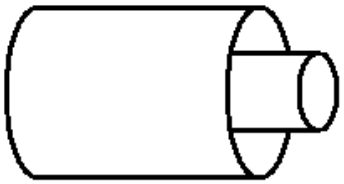
Sensor Arrangements

- We try to make all capacitance measurements differential in order to null out variations in the variable we are trying to measure

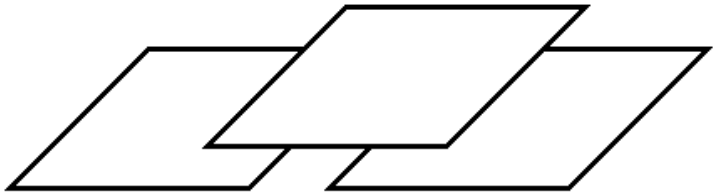
- Arrangements include:



PARALLEL PLATES



COAXIAL CYLINDERS



THREE-PLATE SYSTEMS

Factors to Consider When Measuring Capacitance

- There are many different ways of measuring capacitance. The method used will depend on a number of aspects:

1. Is a differential plate arrangement possible?
2. Are both capacitor plates available to the circuit, or must one be grounded?
3. Is the capacitance large with respect to stray capacitance?
4. What is the bandwidth of measurement?
5. What type of output is desired?
6. Must the circuit work automatically, or can it be manually adjusted?

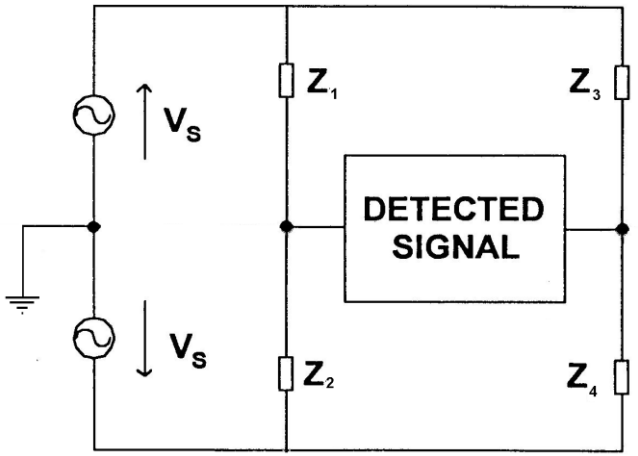
Circuits for Measuring Capacitance

- The circuits we use fall into four basic categories:
  - Bridge Types
  - Resonance Systems
  - Oscillator Systems
  - Charge Pumps

Bridge Circuits

- Used traditionally in systems with differential electrode arrangements
- Considered to be most accurate but require both capacitor plates to be accessible
- If all ratio of impedances is the same size and the voltage sources are the same then the net current flow in the detected signal loop is zero

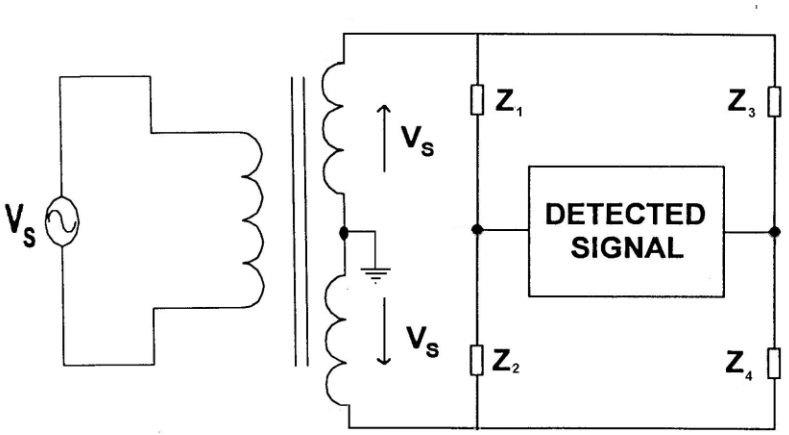
$$\frac{Z_1}{Z_2} = \frac{Z_3}{Z_4}$$



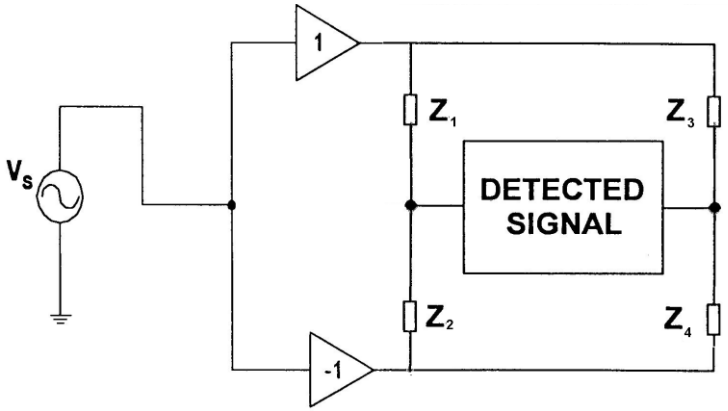
- If all impedances are the same size then no voltage will be developed across the signal detector
- If one of the impedances is different then a voltage will be developed

Bridge Circuits

- Each voltage source must be as well matched as possible
- This is usually implemented with a bifilar-wound transformer
- This arrangement works well up to the limits of the transformer which is between 30kHz – 100kHz



Bridge Circuits



- The transformer can be replaced with power amplifiers

Bridge Circuits

- The detected signal is most easily measured as a voltage

- This has several problems :
  - The relationship between voltage and capacitance is non-linear
  - The system is not stray immune
  - The measurement is degraded if the dielectric is lossy (it dissipates more energy than it should)

Resonance Methods

- Resonance methods work by putting an unknown capacitance in series or parallel with an inductor, so that they form a resonant circuit

- The resonant frequency for a series RLC circuit is:

$$f = \frac{1}{2\pi\sqrt{LC}}$$

- Shifts in the resonant peak can be detected in terms of the resonant frequency, where the amplitude of the current or the phase is:

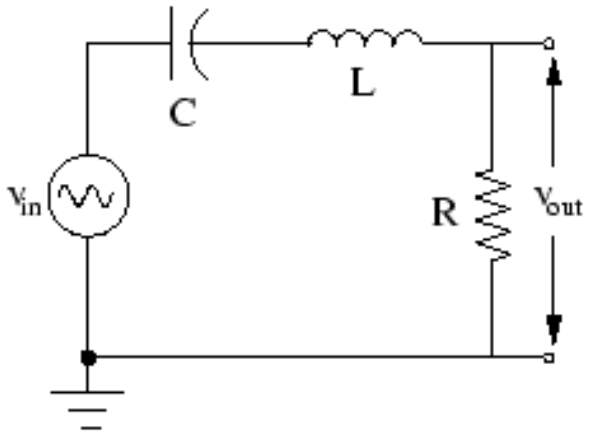
$$i = \frac{v}{R}\left[1 + \frac{i\left(1-\omega^2 LC\right)}{\omega CR}\right]$$

$$\phi = \frac{\left(1-\omega^2 LC\right)}{\omega CR}$$

Resonance Methods

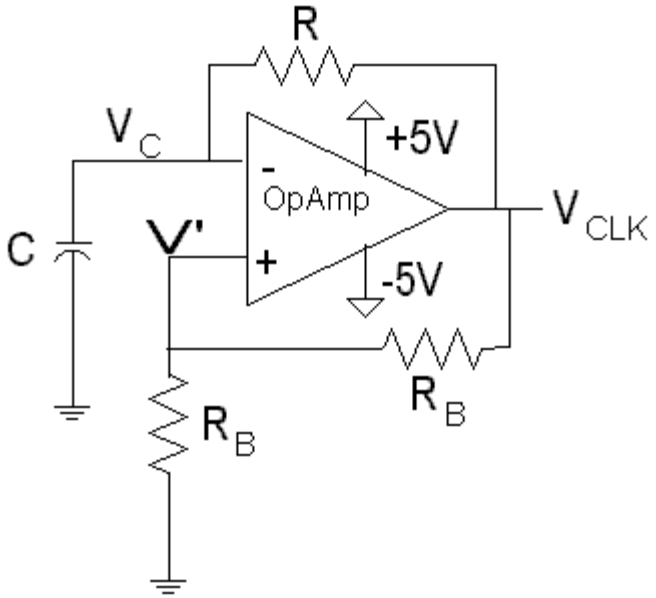
- Advantages include allowing one plate of the capacitor to be grounded and they are very sensitive for detecting shifts in capacitance rather than absolute values of capacitance

- Disadvantages are that they require careful set up in order to avoid stray capacitance and are not normally automatically adjustable



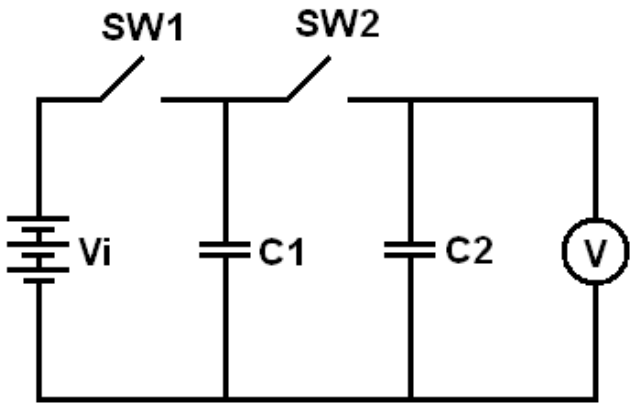
Oscillator Methods

- Use a capacitor as one of the timing elements in an oscillator circuit (Relaxation Oscillator)
- This means a change in capacitance will alter the timing constant of the circuit and therefore the period of the output
- Advantages are that they are simple, automatically adjusting and have a directly accessible output
- Disadvantages are that they are not immune to stray capacitance and are therefore not accurate at very small capacitance values



Charge Pump Techniques

- They are becoming a more common capacitive measurement
- The make use of the principle of comparing a known capacitance to an unknown one by means of charge transfer
- They have the advantage of being stray-immune, automatic and are reasonably good at measuring low values of capacitance
- They give good accuracy in measuring absolute capacitance



$$C_{unknown} = \frac{C_{known}\left(V_i - V_o\right)}{V_o}$$